



Monday Feb-24-2025

Flowserve Facility:

Flowserve Nordstrom / Valbart Valves
1511 Jefferson St.
Sulphur Springs, TX 75482

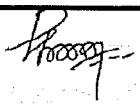
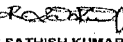
cmtrs@flowserve.com

To:

CERTIFICATE OF CONFORMANCE

SERIAL #	HEAT #	PO #	Description	Class	End Connection
61-618460	1347S	STP46777	4" 2345	900	FLG
61-618461	1347S	STP46777	4" 2345	900	FLG
61-618462	1348S	STP46777	4" 2345	900	FLG
61-618463	1348S	STP46777	4" 2345	900	FLG
61-619325	1358S	STP46777	4" 2345	900	FLG
61-619326	1347S	STP46777	4" 2345	900	FLG
61-619327	1358S	STP46777	4" 2345	900	FLG
61-619328	1369S	STP46777	4" 2345	900	FLG
61-619453	1369S	STP46777	4" 2345	900	FLG
61-619454	1330S	STP46777	4" 2345	900	FLG
61-619455	1330S	STP46777	4" 2345	900	FLG
61-619456	2038S	STP46777	4" 2345	900	FLG
61-619498	2023S	STP46777	4" 2345	900	FLG
61-619499	2019S	STP46777	4" 2345	900	FLG
61-619500	2023S	STP46777	4" 2345	900	FLG

We hereby certify that all the subject valves/parts/sealant have been manufactured in conformance with the applicable requirements of Flowserve Sulphur Springs Operation (FSSO) specifications and that the valve/parts/sealant materials for component parts meet the applicable requirements of FSSO drawings and ASTM specifications, ASME B16.34 (If Applicable), API Q1 10th edition, and API 6D 25th edition.(if applicable to the API monogram). Flowserve Dynamic Balanced Valves are provided with a Pressure balance feature unless specifically identified in the Valve description. All valves/parts/sealant meet all the requirements of the customer purchase order.

	FLOW LINK SYSTEMS (P) LTD FOUNDRY UNIT - II 65 / 2 A, B & C, PUDUPALAYAM B.P.O. AVINASHI TK, COIMBATORE - 641 654 INDIA		TEST CERTIFICATE (EN 10204-3.1)		TC NO : R-4406 DATE : 4-Dec-18								
	CUSTOMER FLOWSERVE INDIA CONTROLS PVT LTD, B8, MMDA Industrial Area, Maraimalai Nagar-603209 Kancheepuram, INDIA		SPECIFICATION REFERENCE 1. ASTM A216-18 Gr.WCC 2. Customer Spec. RMC 90012 Rev. 13		HEAT NO. 2038S Date of Pouring 21-Oct-18 FOUNDRY MARK FLS								
PO NUMBER & DATE 10083548 DT: 25.09.2018			ORDER ACCEPTANCE NO 18-1344										
CHEMICAL COMPOSITION % wt													
MELTING PRACTICE : Steel melted using Electric Induction furnace, and fully killed using Aluminium.													
SPECIFICATION	C	SI	S	P	Mn	Ni	Cr	Mo	V	Cu	Al	RE	CE
Minimum	--	--	--	--	--	--	--	--	--	--	--	--	--
Maximum	0.23	0.60	0.020	0.025	1.20	0.50	0.50	0.20	0.030	0.30	0.08	1.00	0.43
Achieved	0.207	0.429	0.007	0.015	0.939	0.035	0.053	0.009	0.007	0.008	0.042	0.11	0.38
Heat Treatment : NORMALIZING: Temperature raised to 920°C, Soaked for 4 hours and then atmospheric air-cooled													
Visual Inspection - Castings conforms to MSS-SP-55-2011 - Type II to XII (a)													
MECHANICAL PROPERTIES													
TEST TYPE	Tensile as per ASTM A370-17 / ISO 6892-1:2016						Impact (CVN) ISO 148-2016/ASTM A370-17		BEND TEST		HARDNESS		
SPECIFICATION	YS Rp 0.2% (KSI)	YS Rp 1.0% (KSI)	UTS (KSI)	% Elongation		% RA	Test Temperature (at °C) J		ANGLE D		(HBW / HRBW)		
				4d	5d		Values	Avg.					
Minimum	40	--	70	22	--	35	--	--	--	--	--	--	
Maximum	--	--	95	--	--	--	--	--	--	--	--	--	
Achieved	46.5	--	77.4	35	--	65	--	--	--	--	--	152	
DETAIL OF CASTINGS													
S.No.	PO S.No.	DESCRIPTION			CAST DRAWING NO / REV		PART NO		POURED QTY	DESP. QTY			
1	04	4" 900 FE BODY			D-487876 / S		N2757928-60		7	3			
REMARKS: 1. Stated Chemical composition is from heat analysis; and Mechanical properties of test coupon from Cast heel block. 2. *RE = Ni + Cr + Mo + V + Cu 3. ** CE = $C + \frac{Mn}{6} + \frac{Cr+Mo+V}{5} + \frac{Ni+Cu}{15}$ 4. The above stated KSI values have been calculated based on original tested N/MM² values.													
CERTIFIED THAT 1. THE ABOVE GIVEN DETAILS ARE CORRECT, FULFILLS THE SPECIFICATION / STANDARDS AND PO. 2. MATERIAL WAS MANUFACTURED, SAMPLED, TESTED, INSPECTED IN ACCORDANCE WITH THE ABOVE SPECIFICATIONS AND CERTIFIED AS PER EN 10204 - 3.1. 3. THE MATERIAL IS FREE OF CONTAMINATION FROM RADIC ACTIVE ELEMENT OR RADIATION.							MTC - REVIEWED & ACCEPTED  10139336 FICPL QC  M.S.DHANABALAN ASSISTANT MANAGER -QUALITY MANUFACTURER'S AUTHORIZED INSPECTOR						
PREPARED BY: S.MANIBHARATHI			CHECKED BY:  R.SATHISH KUMAR ASSISTANT ENGINEER										